



Introduction to Machine Learning (CS 436/536)

Lecture-3: Mathematical Foundations

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Basic Notations from Calculus (1/4)

| Derivative of $f(x)$ with respect to x

| Partial derivative of a function $f(x,y,\dots)$ with respect to x

– Note: the function may be scalar-valued or vector-valued

- **Derivative of constants.** $\frac{d}{dx}c = 0$.
- **Derivative of linear functions.** $\frac{d}{dx}(ax) = a$.
- **Power rule.** $\frac{d}{dx}x^n = nx^{n-1}$.
- **Derivative of exponentials.** $\frac{d}{dx}e^x = e^x$.
- **Derivative of the logarithm.** $\frac{d}{dx}\log(x) = \frac{1}{x}$.
- **Sum rule.** $\frac{d}{dx}(g(x) + h(x)) = \frac{dg}{dx}(x) + \frac{dh}{dx}(x)$.
- **Product rule.** $\frac{d}{dx}(g(x) \cdot h(x)) = g(x)\frac{dh}{dx}(x) + \frac{dg}{dx}(x)h(x)$.
- **Chain rule.** $\frac{d}{dx}g(h(x)) = \frac{dg}{dh}(h(x)) \cdot \frac{dh}{dx}(x)$.

Basic Notations from Calculus (2/4)

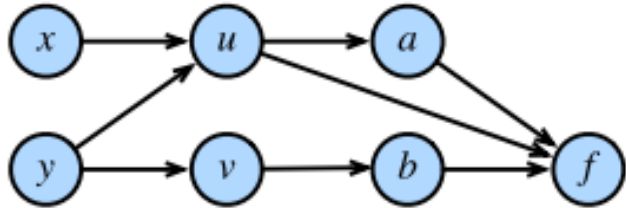
| $\mathfrak{R}^d$: d -dimensional Euclidean space.

| Gradient operator in $\mathfrak{R}^d$: ∇

$$f(x) = 3x^4 - 4x^3 - 12x^2.$$

$$\frac{df}{dx} = 12x^3 - 12x^2 - 24x = 12x(x - 2)(x + 1).$$

Chain Rule (3/4)



$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial a} \frac{\partial a}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial b} \frac{\partial b}{\partial v} \frac{\partial v}{\partial y}.$$

$$f(u, v) = (u + v)^2$$

$$u(a, b) = (a + b)^2, \quad v(a, b) = (a - b)^2,$$

$$a(w, x, y, z) = (w + x + y + z)^2, \quad b(w, x, y, z) = (w + x - y - z)^2.$$

If we want to compute say $\frac{\partial f}{\partial w}$ we may apply the multi-variate chain rule to see:

$$\begin{aligned} \frac{\partial f}{\partial w} &= \frac{\partial f}{\partial u} \frac{\partial u}{\partial w} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial w}, \\ \frac{\partial u}{\partial w} &= \frac{\partial u}{\partial a} \frac{\partial a}{\partial w} + \frac{\partial u}{\partial b} \frac{\partial b}{\partial w}, \\ \frac{\partial v}{\partial w} &= \frac{\partial v}{\partial a} \frac{\partial a}{\partial w} + \frac{\partial v}{\partial b} \frac{\partial b}{\partial w}. \end{aligned}$$

Given f is function of u and v , so if we want to calculate the $\frac{\partial f}{\partial w}$ then we must use the chain rule:

Suppose, f is only function of u then $\frac{\partial f}{\partial w}$ can be written as follows:

$$\frac{\partial f}{\partial w} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial w} \Rightarrow \frac{\partial f}{\partial w}$$

We have done this because we have a way to calculate $\frac{\partial f}{\partial u}$, because f is a function of u .

Now, in the previous example given that f is function of u and v , so applying multi-variate chain rule.

$$\frac{\partial f}{\partial w} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial w} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial w}$$

Now, to calculate $\frac{\partial u}{\partial w}$, suppose, “ u ” is only function of “ a ” then $\frac{\partial u}{\partial w}$ can be written as follows:

$$\frac{\partial u}{\partial w} = \frac{\partial u}{\partial a} \cdot \frac{\partial a}{\partial w} \Rightarrow \frac{\partial u}{\partial w}$$

Now, in the previous example given that u is function of “ a ” and “ b ”, so applying multi-variate chain rule.

$$\frac{\partial u}{\partial w} = \frac{\partial u}{\partial a} \cdot \frac{\partial a}{\partial w} + \frac{\partial u}{\partial b} \cdot \frac{\partial b}{\partial w} \quad \text{Similarly} \Rightarrow \frac{\partial v}{\partial w} = \frac{\partial v}{\partial a} \cdot \frac{\partial a}{\partial w} + \frac{\partial v}{\partial b} \cdot \frac{\partial b}{\partial w}$$

Now “ a ” and “ b ” both are function of w , therefore $\frac{\partial a}{\partial w}$ and $\frac{\partial b}{\partial w}$ can be calculated by simple differentiation.

Working Example: Chain Rule

$$f(x) = (4x^5 - 1)\sqrt[3]{x + 1}$$

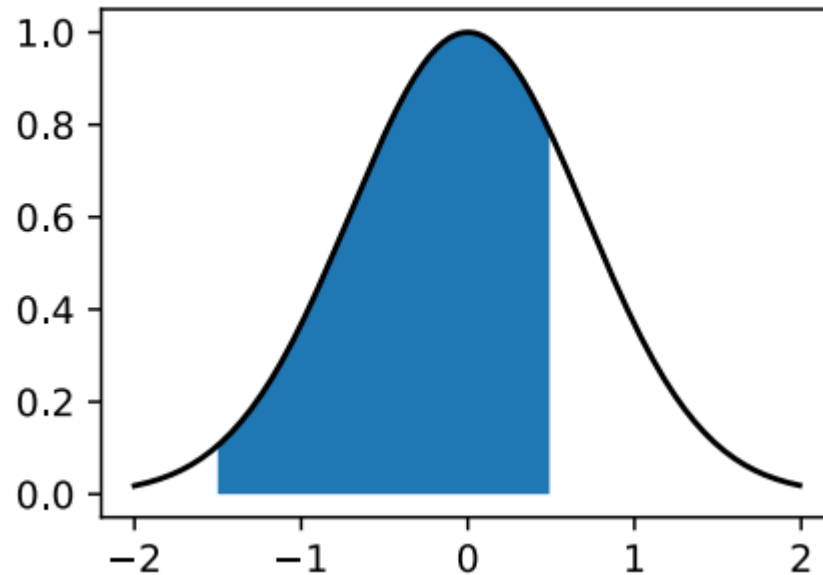
Working Example: Chain Rule

$$f(x) = (4x^5 - 1)\sqrt[3]{x + 1}$$

$$f'(x) = (4x^5 - 1) \cdot \frac{1}{3}(x + 1)^{-\frac{2}{3}} + (x + 1)^{\frac{1}{3}} \cdot 20x^4$$

Basic Notations from Calculus (4/4)

| The integral of $f(x)$ between a and b



$$\text{Area}(\mathcal{A}) = \int_a^b f(x) \, dx.$$

Basic Notations from Set Theory (1/2)

| **A set S is a collection of objects.**

- $\emptyset$: the empty set (a special set that contains no object)

| **Some basic relations and operations**

- $x \in A$: An object x is a member of a set A .
- $A \subseteq B$: Set A is a *subset* of $B \iff x \in A \Rightarrow x \in B$
- $B \subset C$: Set B is a *proper subset* of C .

Basic Notations from Set Theory (2/2)

| Some basic relations and operations

- $A \cup B$: The union of A and B .
- $A \cap B$: The intersection of A and B . (AB in shorthand)
- A^c or $\overline{A}$: The complement of A
- A and B are disjoint if $A \cap B = \emptyset$

Linear Algebra: Basic Notations (1/2)

| A d -dimensional column vector $\mathbf{x}$ and its transpose $\mathbf{x}^t$

| n by d matrix M and its d by n transpose M^t

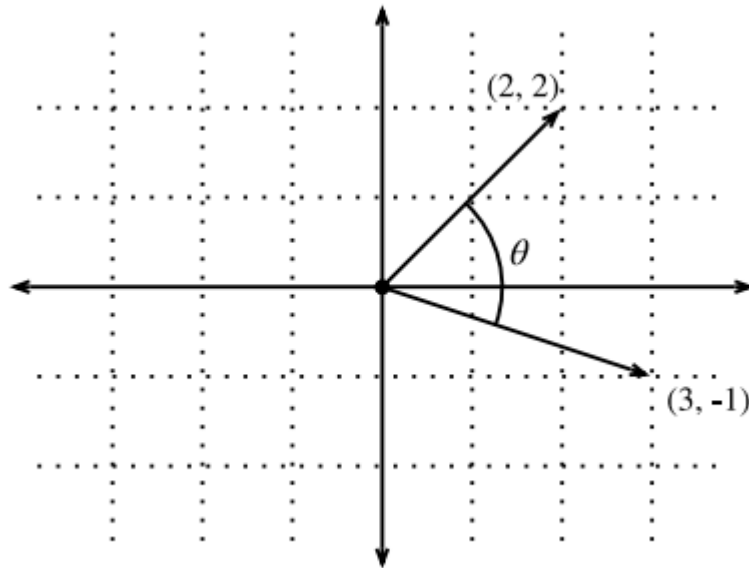
$$\mathbf{x} = \begin{bmatrix} 1 \\ 7 \\ 0 \\ 1 \end{bmatrix},$$

$$\mathbf{x}^T = [1 \quad 7 \quad 0 \quad 1].$$

Linear Algebra: Basic Notations (1/2)

| The length or Euclidean norm of a vector $\mathbf{x}$,
denoted $\|\mathbf{x}\|$

| Normalized vector, $\|\mathbf{x}\| = 1$



$$\theta = \arccos \left(\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} \right) .$$

Working Example: Vector and Angle

What is the length of the two vectors and the angle between them?

Suppose these two vectors as $\vec{u}$ and $\vec{v}$, to calculate angle between them we calculate their scalar length first.

$$|u| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \Rightarrow \sqrt{(2 - 0)^2 + (2 - 0)^2} \Rightarrow 2\sqrt{2}$$

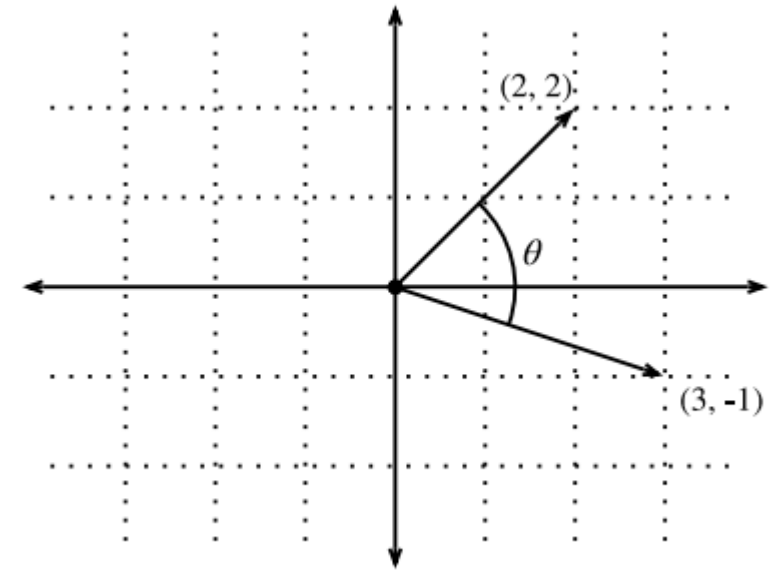
$$|v| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \Rightarrow \sqrt{(3 - 0)^2 + (-1 - 0)^2} \Rightarrow \sqrt{10}$$

Now we calculate the dot product of both vectors:

$$u \cdot v = \langle 2, 2 \rangle \cdot \langle 3, -1 \rangle \Rightarrow (2 \times 3) + (2 \times -1) \Rightarrow 4$$

Now to calculate the angle between them we use the formula:

$$\theta = \cos^{-1} \frac{4}{(2\sqrt{2})(\sqrt{10})} \Rightarrow 1.107 \text{ Radian or } \mathbf{63.43 \text{ Degrees}}$$



Matrix: Additional Definitions

| Matrix inversion $\mathbf{A}^{-1}$

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$\frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

| Eigenvectors and Eigenvalues of $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}.$$

If we consider $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$, we see this is equivalent to the polynomial equation $0 = (2 - \lambda)(3 - \lambda) - 2 = (4 - \lambda)(1 - \lambda)$. Thus, two eigenvalues are 4 and 1. To find the associated vectors, we then need to solve

$$\begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \end{bmatrix}.$$

We can solve this with the vectors $[1, -1]^\top$ and $[1, 2]^\top$ respectively.

Let us continue the previous example one step further. Let

$$\mathbf{W} = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix},$$

be the matrix where the columns are the eigenvectors of the matrix $\mathbf{A}$. Let

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix},$$

$$\mathbf{A} = \mathbf{W}\Sigma\mathbf{W}^{-1}.$$

$$\mathbf{A}^n = \overbrace{\mathbf{A} \cdots \mathbf{A}}^{n \text{ times}} = \overbrace{(\mathbf{W}\Sigma\mathbf{W}^{-1}) \cdots (\mathbf{W}\Sigma\mathbf{W}^{-1})}^{n \text{ times}} = \mathbf{W} \overbrace{\Sigma \cdots \Sigma}^{n \text{ times}} \mathbf{W}^{-1} = \mathbf{W}\Sigma^n\mathbf{W}^{-1}.$$

Determinant

In general, if we have a matrix

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

We define $ad - bc$ as the determinant.

Determinant

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$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

we can see with some computation that the area of the resulting parallelogram is $ad - bc$. This area is referred to as the determinant.

$$\mathbf{B} = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}.$$

Determinant

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we can see with some computation that the area of the resulting parallelogram is $ad - bc$. This area is referred to as the determinant.

$$\mathbf{B} = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix} \Rightarrow ? \quad 0$$

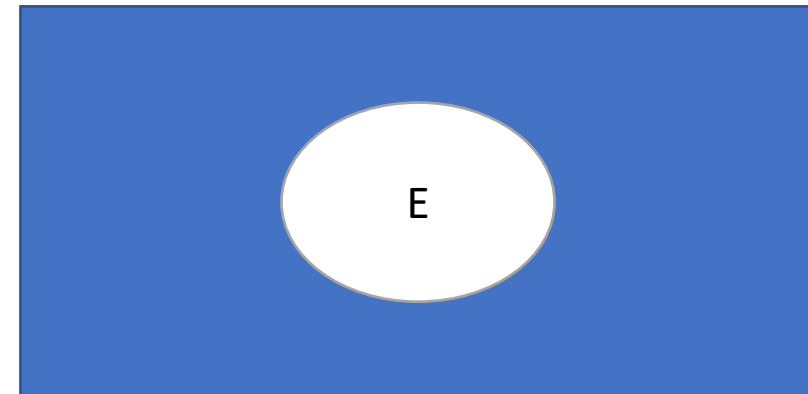
Some key points

- | The eigen decomposition of a matrix can allow for many operations to be reduced to operations on the eigenvalues.
- | Vectors can be interpreted geometrically as either points or directions in space.
- | Dot products define the notion of angle to arbitrarily high-dimensional spaces.
- | The rank of a matrix is the size of the largest subset of its columns that are linearly independent.
- | The rank is the same as the number of non-zero eigenvalues of A .
- | When a matrix's inverse is defined, matrix inversion allows us to find another matrix that undoes the action of the first.
- | Matrix inversion is useful in theory, but requires care in practice owing to numerical instability.

Probability Space (1/2)

| A probability space is a triplet $(\Omega, \mathcal{B}, P)$ that is used to model a process or an experiment with random outcomes.

- The **sample space** Ω is the set of all possible outcomes of an experiment
 - Consider two different experiments
 - (1) Tossing a coin; (2) Tossing a die



Probability Space (2/2)

| $\mathcal{B}$: a sigma algebra (or Borel field), or informally, a collection of subsets of Ω , subject to some constraints (like containing the empty set, being closed under complements and countable union)

| P : a measure called probability defined on $\mathcal{B}$, that satisfies

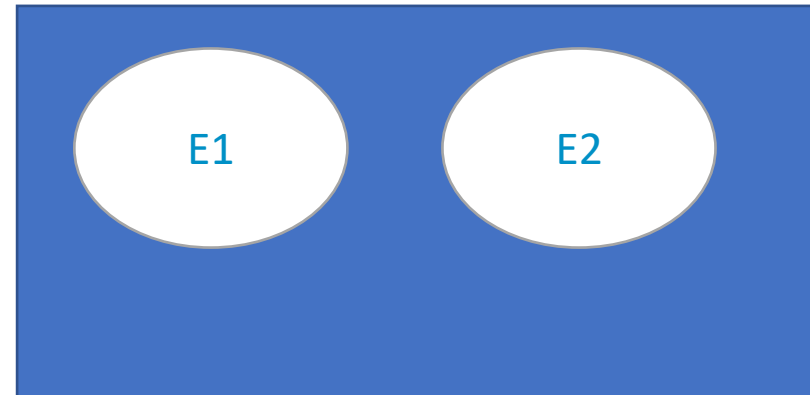
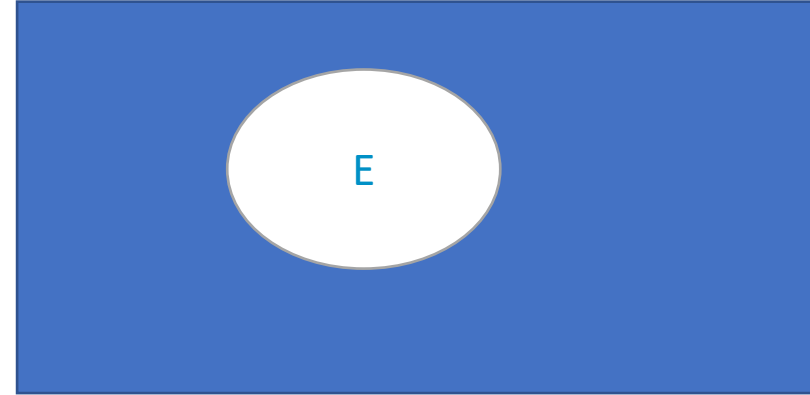
- $P(A) \geq 0$ for all $A \in \mathcal{B}$
- $P(\Omega) = 1$
- If $A_1, A_2, \dots \in \mathcal{B}$ are pairwise disjoint then $P(\cup A_i) = \sum P(A_i)$
(i.e., $A_j \cap A_k = \emptyset, \forall j \neq k$)
- Please review the cheatsheet: [CME 106 - Probability Cheatsheet \(stanford.edu\)](https://cme106.stanford.edu/cheatsheet/)

Probability Axioms

$$| 1 \geq P(E) \geq 0$$

$$| P(S) = 1$$

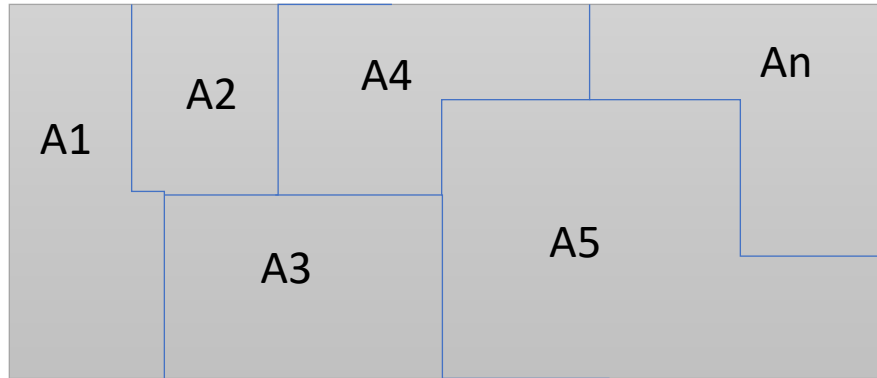
$$| P(E1 \cup E2) = P(E1) + P(E2)$$



Bayes' Theorem

For events A and B such that $P(B) > 0$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$



For Any event B in the sample space
above extended bayes' theorem:

$$P(A_k|B) = \frac{P(B|A_k)P(A_k)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

Working Example: Bayes Examples (1/2)

Probability	Red	Not Red	Total
Man	5	55	60
Woman	30	10	40
Total	35	65	100

What is the probability that a person wearing red is a man? Or $P(\text{Man}|\text{red}) = ?$

Here, we need to calculate the probability of a person being a Man given that the person is wearing Red. Therefore, total number of Red wearers is 35 and total number of men wearing Red is 5. Hence,

$$P(\text{Man}|\text{Red}) = \frac{5}{35}$$

Let's try this again using Bayes' Theorem:

$$P(\text{Man}|\text{Red}) = \frac{P(\text{Red}|\text{Man}) \cdot P(\text{Man})}{P(\text{Red})}$$

$P(\text{Red}|\text{Man}) = \text{Probability of a person wearing red given that they are a Man}$

$$P(\text{Red}|\text{Man}) = \frac{5}{60} \quad \text{and} \quad P(\text{Man}) = \frac{60}{100} \quad \text{and} \quad P(\text{Red}) = \frac{35}{100}$$

$$P(\text{Man}|\text{Red}) = \frac{P(\text{Red}|\text{Man}) \cdot P(\text{Man})}{P(\text{Red})} \Rightarrow \frac{(\frac{5}{60})(\frac{60}{100})}{\frac{35}{100}} \Rightarrow \frac{5}{35}$$

We are getting same result by Bayes' Theorem as well.

What is the probability that a man wears red? Or $P(\text{Red}|\text{Man}) = ?$

This is same as we calculated in previous step:

$$P(\text{Red}|\text{Man}) = \frac{5}{60}$$

Working Example: Bayes Examples (2/2)

- Sam put in 15 paintings, 4% of his works have won First Prize.
- Jon put in 5 paintings, 6% of his works have won First Prize.
- Pablo put in 10 paintings, 3% of his works have won First Prize.

What is the chance that Sam will win First Prize in this competition?

$$P(\text{Sam}|\text{First}) = \frac{P(\text{Sam})P(\text{First}|\text{Sam})}{P(\text{Sam})P(\text{First}|\text{Sam}) + P(\text{Jon})P(\text{First}|\text{Jon}) + P(\text{Pablo})P(\text{First}|\text{Pablo})}$$

$P(\text{Sam})$ = Probability of a painting being Sam's painting = (Sam's Paintings/Total) = 15/30

$P(\text{First}|\text{Sam})$ = Probability of a painting winning first price given it is submitted by Sam
=> 4% of Sam's Total work => 4/100 .

$P(\text{Jon}) = \text{Probability of a painting being Jon's painting} = (\text{Jon's Paintings}/\text{Total}) = 5/30$

$P(\text{First}|\text{Jon}) = \text{Probability of a painting winning first price given it is submitted by Jon}$
 $\Rightarrow 6\% \text{ of Jon's Total work} \Rightarrow 6/100 .$

$P(\text{Pablo}) = \text{Probability of a painting being Pablo's painting} = (\text{Pablo's Paintings}/\text{Total}) = 10/30$

$P(\text{First}|\text{Pablo}) = \text{Probability of a painting winning first price given it is submitted by Pablo}$
 $\Rightarrow 3\% \text{ of Pablo's Total work} \Rightarrow 3/100 .$

$$P(\text{Sam}|\text{First}) = \frac{\left(\frac{15}{30}\right)\left(\frac{4}{100}\right)}{\left(\frac{15}{30}\right)\left(\frac{4}{100}\right) + \left(\frac{5}{30}\right)\left(\frac{6}{100}\right) + \left(\frac{10}{30}\right)\left(\frac{3}{100}\right)} \Rightarrow \frac{60}{60+30+30} \Rightarrow 0.5 \Rightarrow 50\% \text{ Chance}$$

Working Example: Bayes Examples (2/3)

- Let us say $P(\text{Fire})$ means how often there is fire, and $P(\text{Smoke})$ means how often we see smoke, then:
- $P(\text{Fire}|\text{Smoke})$ means how often there is fire when we can see smoke
- $P(\text{Smoke}|\text{Fire})$ means how often we can see smoke when there is fire
- fires are rare (1%) but smoke is common (10%) due to barbecues,
- and 90% of dangerous fires make smoke
- We can then discover the probability of dangerous Fire when there is Smoke:

$$P(\text{Fire}) = 1\% \Rightarrow 1/100$$

$$P(\text{Smoke}|\text{Fire}) = 90\% \Rightarrow 90/100$$

$$P(\text{Smoke}) = 10\% \Rightarrow 10/100$$

$$P(\text{Fire}|\text{Smoke}) = \frac{P(\text{Fire}) P(\text{Smoke}|\text{Fire})}{P(\text{Smoke})}$$

$$\Rightarrow \frac{\left(\frac{1}{100}\right)\left(\frac{90}{100}\right)}{\frac{10}{100}} \Rightarrow \frac{9}{100} \Rightarrow 9\%$$

So, it is still worth checking out any smoke to be sure.

Probability

Joint Probability: $P(A = a, B = b)$

Conditional Probability: $P(B = b \mid A = a):$

Sum Rule: $P(B) = \sum_A P(A, B),$

An Example:

- $P(H=1) = 0.0015$
- $P(H=1 | D=1) = ?$

Conditional Probability	H=1	H=0
$P(D=1 H)$	1	0.01
$P(D=0 H)$	0	0.99

An Example:

- $P(H=1) = 0.0015$
- $P(H=1|D=1) = ?$

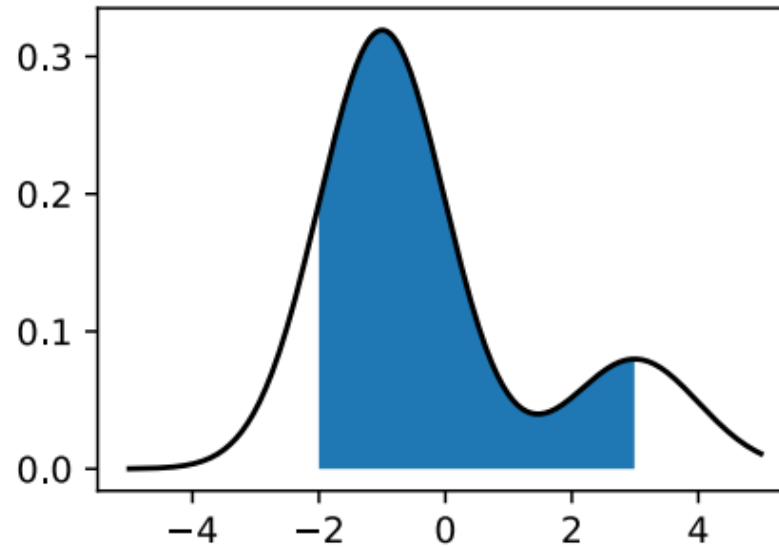
$$\begin{aligned} &P(D_1 = 1) \\ &= P(D_1 = 1, H = 0) + P(D_1 = 1, H = 1) \\ &= P(D_1 = 1 | H = 0)P(H = 0) + P(D_1 = 1 | H = 1)P(H = 1) \\ &= 0.011485. \end{aligned}$$

Thus, we get

$$\begin{aligned} &P(H = 1 | D_1 = 1) \\ &= \frac{P(D_1 = 1 | H = 1)P(H = 1)}{P(D_1 = 1)}. \\ &= 0.1306 \end{aligned}$$

Conditional Probability	H=1	H=0
$P(D=1 H)$	1	0.01
$P(D=0 H)$	0	0.99

Probability Distribution Function:



$$\int_{-\infty}^{\infty} p(x) \, dx = 1.$$

$$P(X \in (a, b]) = \int_a^b p(x) \, dx.$$

Cumulative Distribution Function:

$$F(x) = \int_{-\infty}^x p(x) \, dx = P(X \leq x).$$

$$F(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{2} & 0 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

- $F(x) \rightarrow 0$ as $x \rightarrow -\infty$.
- $F(x) \rightarrow 1$ as $x \rightarrow \infty$.
- $F(x)$ is non-decreasing ($y > x \implies F(y) \geq F(x)$).
- $F(x)$ is continuous (has no jumps) if X is a continuous random variable.

Mean and Variance:

| Most commonly used statistics are Mean and Variance:

Mean: $\mu_X = E[X] = \sum_i x_i p_i.$

As a minimalistic example that we will examine throughout this section, let us take X to be the random variable which takes the value $a - 2$ with probability p , $a + 2$ with probability p and a with probability $1 - 2p$.

$$\mu_X = E[X] = \sum_i x_i p_i = (a - 2)p + a(1 - 2p) + (a + 2)p = a.$$

Because they are helpful, let us summarize a few properties. • For any random variable X and numbers a and b , we have that $\mu(aX+b) = a\mu_X + b$. • If we have two random variables X and Y , we have $\mu(X+Y) = \mu_X + \mu_Y$.

Mean and Variance:

| If we look at the typical size of this quantity by taking the mean, we arrive at the variance:

$$\sigma_X^2 = \text{Var}(X) = E[(X - \mu_X)^2] = E[X^2] - \mu_X^2.$$

$$E[X^2] = (a - 2)^2 p + a^2(1 - 2p) + (a + 2)^2 p = a^2 + 8p.$$

$$\sigma_X^2 = \text{Var}(X) = E[X^2] - \mu_X^2 = a^2 + 8p - a^2 = 8p.$$

Mean and Variance:

| For any random variable X , $\text{Var}(X) \geq 0$, with $\text{Var}(X) = 0$ if and only if X is a constant.

| For any random variable X and numbers a and b , we have that $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

| If we have two independent random variables X and Y , we have $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Standard Deviation:

| This summary statistics can always be deduced from the variance by taking the square root!

| Thus we define the standard deviation to be

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

Standard Deviation:

| In our example, this means we now have the standard deviation is $\sigma = \sqrt{2p}$.

| The properties we had for the variance can be restated for the standard deviation.

| For any random variable X , $\sigma \geq 0$.

| For any random variable X and numbers a and b , we have that $\sigma(aX+b) = |a|\sigma X$

| If we have two independent random variables X and Y , we have $\sigma(X+Y) = \sqrt{(\sigma^2 X + \sigma^2 Y)}$.

Joint Density Function and Marginal Distribution

| **For some function $p(x, y)$. This is referred to as the joint density of X and Y . Similar properties are true for this as we saw in the single variable case. Namely:**

- $p(x, y) \geq 0$;
- $\int_{\mathbb{R}^2} p(x, y) \, dx \, dy = 1$;
- $P((X, Y) \in \mathcal{D}) = \int_{\mathcal{D}} p(x, y) \, dx \, dy$.

$$\begin{aligned} p_X(x) &\approx \sum_i \epsilon p_{X,Y}(x, \epsilon \cdot i) \\ &\approx \int_{-\infty}^{\infty} p_{X,Y}(x, y) \, dy. \end{aligned}$$

$$p_X(x) = \int_{-\infty}^{\infty} p_{X,Y}(x, y) \, dy.$$

Marginal Distribution

$$\sigma_{XY} = \text{Cov}(X, Y) = \sum_{i,j} (x_i - \mu_X)(y_j - \mu_Y)p_{ij} = E[XY] - E[X]E[Y].$$

$$\sigma_{XY} = \int_{\mathbb{R}^2} (x - \mu_X)(y - \mu_Y)p(x, y) \, dx \, dy.$$

$$P(X = 1 \text{ and } Y = -1) = \frac{p}{2},$$

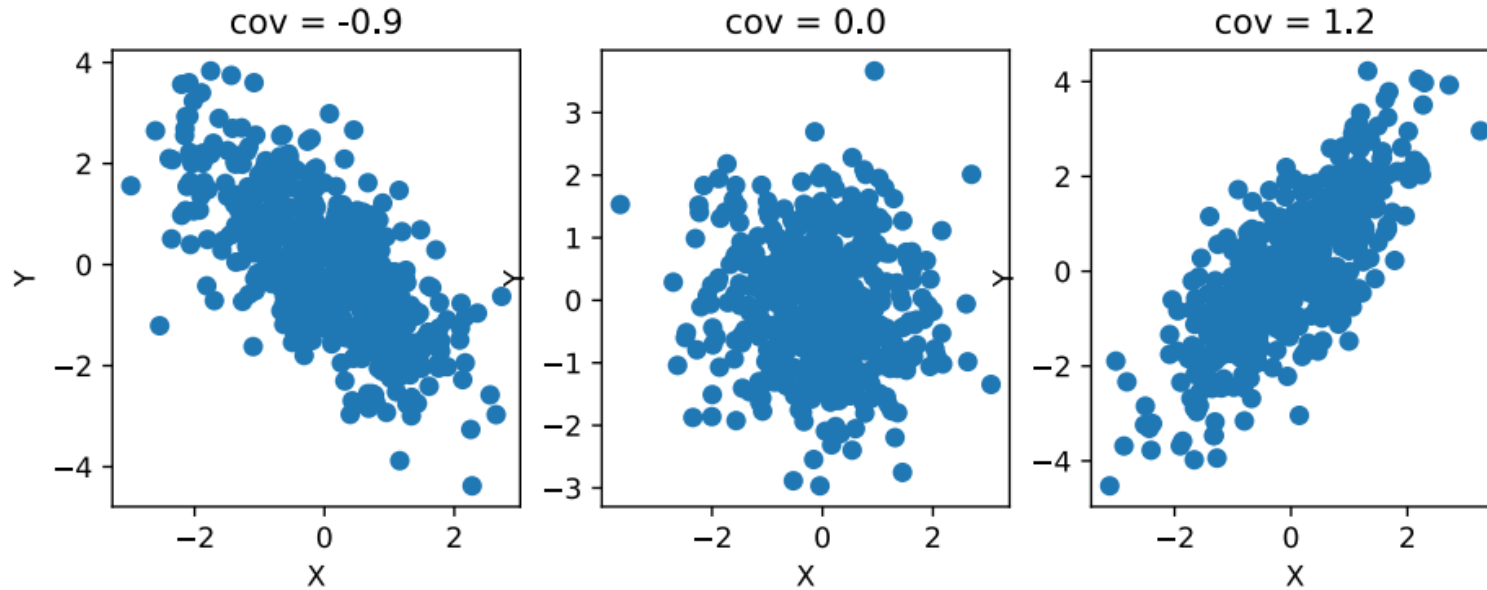
$$P(X = 1 \text{ and } Y = 3) = \frac{1-p}{2},$$

$$P(X = 3 \text{ and } Y = -1) = \frac{1-p}{2},$$

$$P(X = 3 \text{ and } Y = 3) = \frac{p}{2},$$

$$\begin{aligned} \text{Cov}(X, Y) &= \sum_{i,j} (x_i - \mu_X)(y_j - \mu_Y)p_{ij} \\ &= (1-2)(-1-1)\frac{p}{2} + (1-2)(3-1)\frac{1-p}{2} + (3-2)(-1-1)\frac{1-p}{2} + (3-2)(3-1)\frac{p}{2} \\ &= 4p - 2. \end{aligned}$$

Covariance Properties



Let us see some properties of covariances:

- For any random variable X , $\text{Cov}(X, X) = \text{Var}(X)$.
- For any random variables X, Y and numbers a and b , $\text{Cov}(aX + b, Y) = \text{Cov}(X, aY + b) = a\text{Cov}(X, Y)$.
- If X and Y are independent then $\text{Cov}(X, Y) = 0$.

Correlation

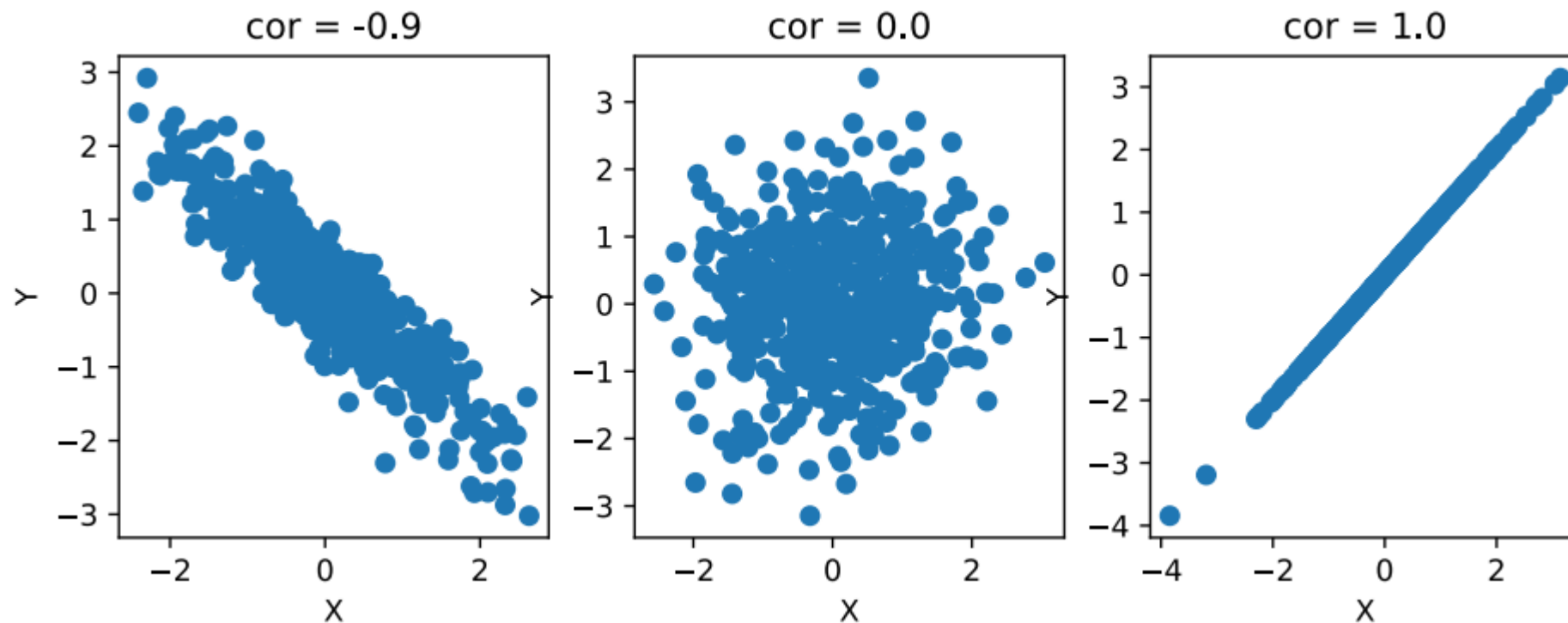
$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y},$$

$$\rho(X, Y) = \frac{4p - 2}{1 \cdot 2} = 2p - 1.$$

Let us list a few properties of the correlation below.

- For any random variable X , $\rho(X, X) = 1$.
- For any random variables X, Y and numbers a and b , $\rho(aX + b, Y) = \rho(X, aY + b) = \rho(X, Y)$.
- If X and Y are independent with non-zero variance then $\rho(X, Y) = 0$.

Correlation



$$\cos(\theta) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} = \frac{\sum_i v_i w_i}{\sqrt{\sum_i v_i^2} \sqrt{\sum_i w_i^2}}.$$

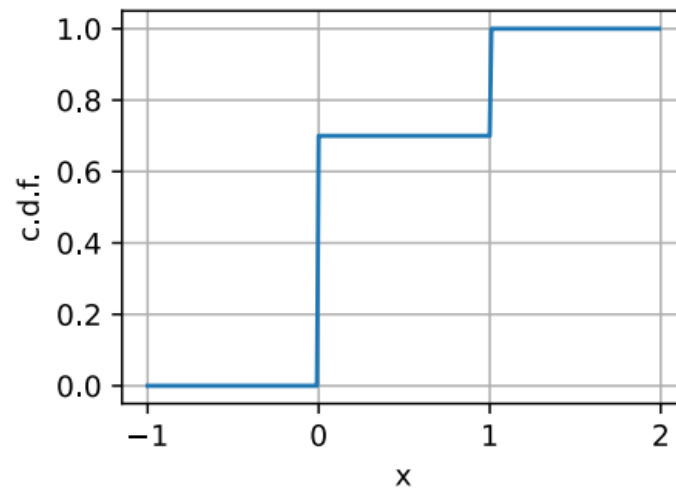
Summary

- ❑ The probability density function allows us to work with continuous random variables by giving a function where the area under the curve on some interval gives the probability of finding a sample point in that interval.
- ❑ The cumulative distribution function is the probability of observing the random variable to be less than a given threshold.
- ❑ The mean is the average value of a random variable.
- ❑ The variance is the expected square of the difference between the random variable and its mean.
- ❑ The standard deviation is the square root of the variance. It can be thought of as measuring the range of values the random variable may take.
- ❑ Joint densities allow us to work with correlated random variables.
- ❑ We may marginalize joint densities by integrating over unwanted random variables to get the distribution of the desired random variable.
- ❑ The covariance and correlation coefficient provide a way to measure any linear relationship between two correlated random variables.

Distribution: Bernoulli

$$X \sim \text{Bernoulli}(p).$$

The cumulative distribution function is



$$F(x) = \begin{cases} 0 & x < 0, \\ 1 - p & 0 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

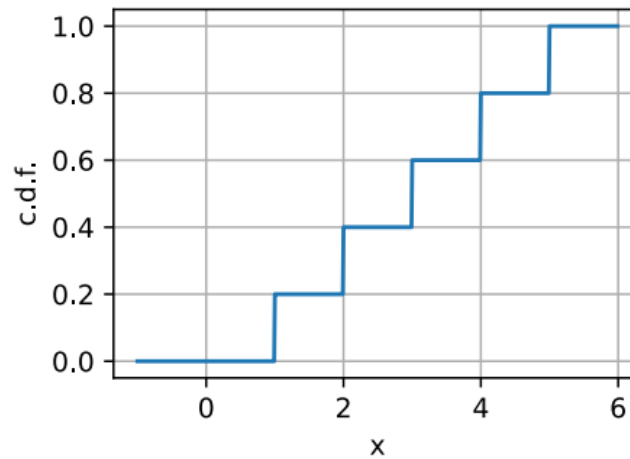
If $X \sim \text{Bernoulli}(p)$, then:

- $\mu_X = p$,
- $\sigma_X^2 = p(1 - p)$.

Distribution: Discrete Uniform

$$X \sim U(n).$$

The cumulative distribution function is



$$F(x) = \begin{cases} 0 & x < 1, \\ \frac{k}{n} & k \leq x < k+1 \text{ with } 1 \leq k < n, \\ 1 & x \geq n. \end{cases}$$

If $X \sim U(n)$, then:

- $\mu_X = \frac{1+n}{2},$
- $\sigma_X^2 = \frac{n^2-1}{12}.$

Distribution: Continuous Uniform

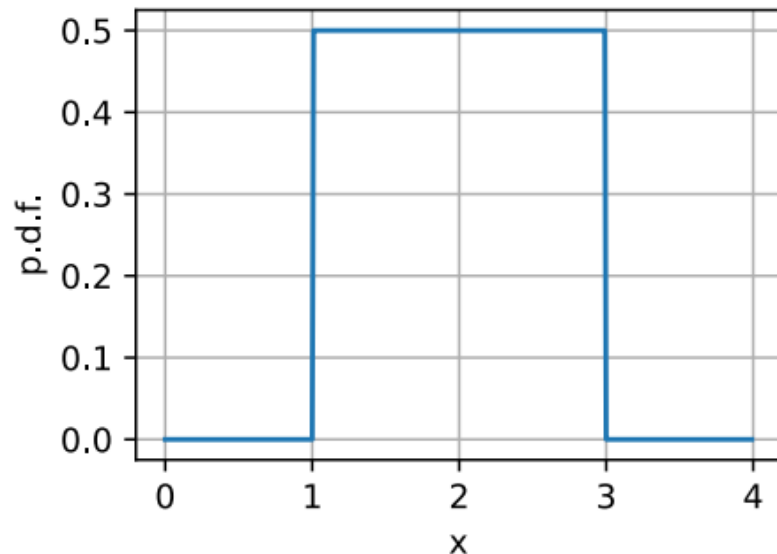
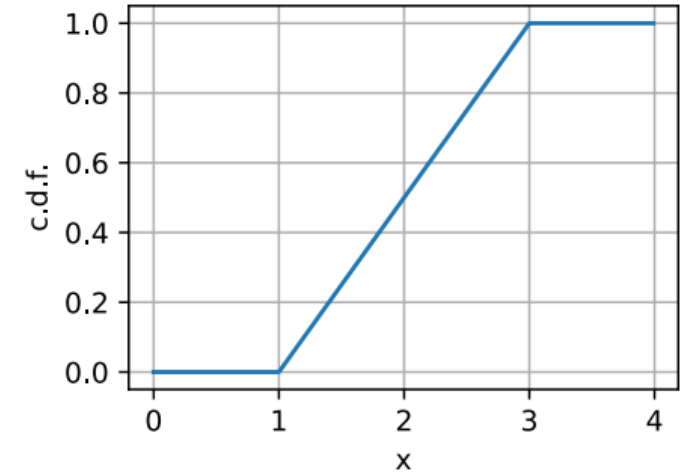
$$X \sim U(a, b).$$

The probability density function is

$$p(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b], \\ 0 & x \notin [a, b]. \end{cases}$$

The cumulative distribution function is

$$F(x) = \begin{cases} 0 & x < a, \\ \frac{x-a}{b-a} & x \in [a, b], \\ 1 & x \geq b. \end{cases}$$



If $X \sim U(a, b)$, then:

- $\mu_X = \frac{a+b}{2},$
- $\sigma_X^2 = \frac{(b-a)^2}{12}.$

Distribution: Binomial

$$X = \sum_{i=1}^n X_i.$$

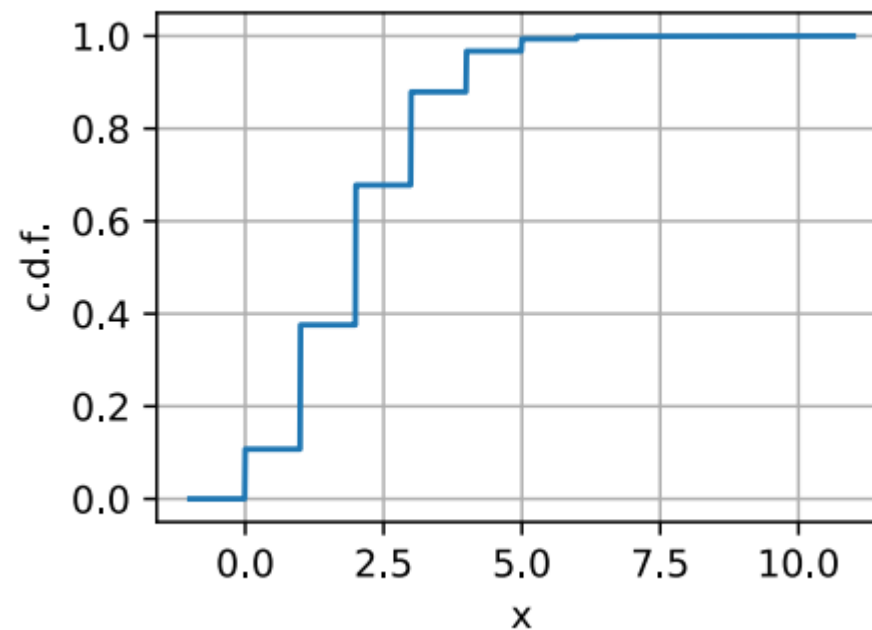
In this case, we will write

$$X \sim \text{Binomial}(n, p).$$

$$F(x) = \begin{cases} 0 & x < 0, \\ \sum_{m \leq k} \binom{n}{m} p^m (1-p)^{n-m} & k \leq x < k+1 \text{ with } 0 \leq k < n, \\ 1 & x \geq n. \end{cases}$$

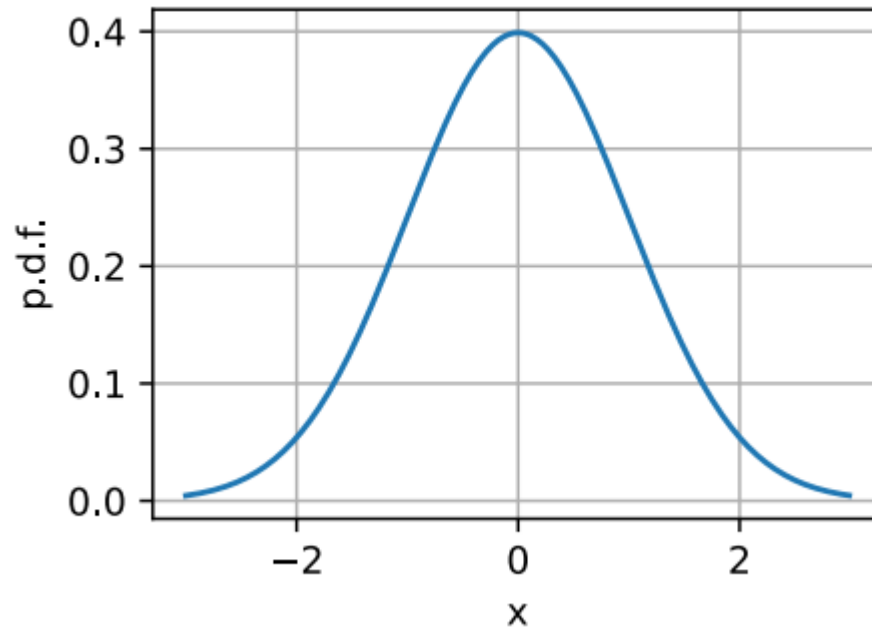
If $X \sim \text{Binomial}(n, p)$, then:

- $\mu_X = np$,
- $\sigma_X^2 = np(1-p)$.



Distribution: Gaussian

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$



- $\mu_X = \mu,$
- $\sigma_X^2 = \sigma^2.$

Entropy Motivation

- The information we gain by observing a random variable does not depend on what we call the elements, or the presence of additional elements which have probability zero.
- The information we gain by observing two random variables is no more than the sum of the information we gain by observing them separately. If they are independent, then it is exactly the sum.
- The information gained when observing (nearly) certain events is (nearly) zero.

Entropy:

- For any random variable X that follows a probability distribution P with a probability density function (p.d.f.) or a probability mass function (p.m.f.) $p(x)$
- We measure the expected amount of information through entropy (or Shannon entropy)

$$H(X) = -E_{x \sim P}[\log p(x)].$$

To be specific, if X is discrete,

$$H(X) = -\sum_i p_i \log p_i, \text{ where } p_i = P(X_i).$$

Otherwise, if X is continuous, we also refer entropy as *differential entropy*

$$H(X) = -\int_x p(x) \log p(x) \, dx.$$

Working Example: Entropy

$$P(X = \textit{heads}) = \frac{1}{2} \qquad P(X = \textit{tails}) = \frac{1}{2}.$$

$$\begin{aligned} H(P) &= - \sum_{x \in \{\textit{heads}, \textit{tails}\}} P(x) \log P(x) \\ &= - \left[\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} \right] \\ &= - \left[-\frac{1}{2} + -\frac{1}{2} \right] \\ &= 1. \end{aligned}$$

Entropy Intuition

- Why do we use an expectation of a negative logarithm?
- Why Log?

Suppose that $p(x) = f_1(x)f_2(x) \dots, f_n(x)$, where each component function $f_i(x)$ is independent from each other. This means that each $f_i(x)$ contributes independently to the total information obtained from $p(x)$.

- Why Negative?

Intuitively, more frequent events should contain less information than less common events, since we often gain more information from an unusual case than from an ordinary one. However, log is monotonically increasing with the probabilities, and indeed negative for all values in $[0, 1]$.

Maximum Likelihood Example

Suppose that we have a single parameter θ representing the probability that a coin flip is heads. Then the probability of getting a tails is $1-\theta$, and so if our observed data X is a sequence with n_H heads and n_T tails, we can use the fact that independent probabilities multiply to see that

$$P(X | \theta) = \theta^{n_H} (1 - \theta)^{n_T}.$$

If we flip 13 coins and get the sequence “HHHTHTTHHHHT”, which has $n_H = 9$ and $n_T = 4$, we see that this is

$$P(X | \theta) = \theta^9 (1 - \theta)^4.$$

$$\begin{aligned} 0 &= \frac{d}{d\theta} P(X | \theta) \\ &= \frac{d}{d\theta} \theta^9 (1 - \theta)^4 \\ &= 9\theta^8 (1 - \theta)^4 - 4\theta^9 (1 - \theta) \\ &= \theta^8 (1 - \theta)^3 (9 - 13\theta). \end{aligned}$$

This has three solutions: 0, 1 and 9/13. The first two are clearly minima, not maxima as they assign probability 0 to our sequence. The final value does not assign zero probability to our sequence, and thus must be the maximum likelihood estimate $\hat{\theta} = 9/13$

Negative Log Likelihood

$$(1/2)^{1000000000} \qquad \log((1/2)^{1000000000}) = 1000000000 \cdot \log(1/2) \approx -301029995.6 \dots$$

i) Precision

$$\log(P(X \mid \theta)).$$

ii) The logarithm turns products to sums

We often work with loss functions, where we wish to minimize the loss. We may turn maximum likelihood into the minimization of a loss by taking $-\log(P(X \mid \theta))$, which is the negative loglikelihood.

$$-\log(P(X \mid \theta)) = -\log(\theta^{n_H} (1 - \theta)^{n_T}) = -(n_H \log(\theta) + n_T \log(1 - \theta)).$$

Cross-Entropy as An Objective Function of Multi-class Classification:

Assume that our model is parameterized by θ . For true label vectors y_i .

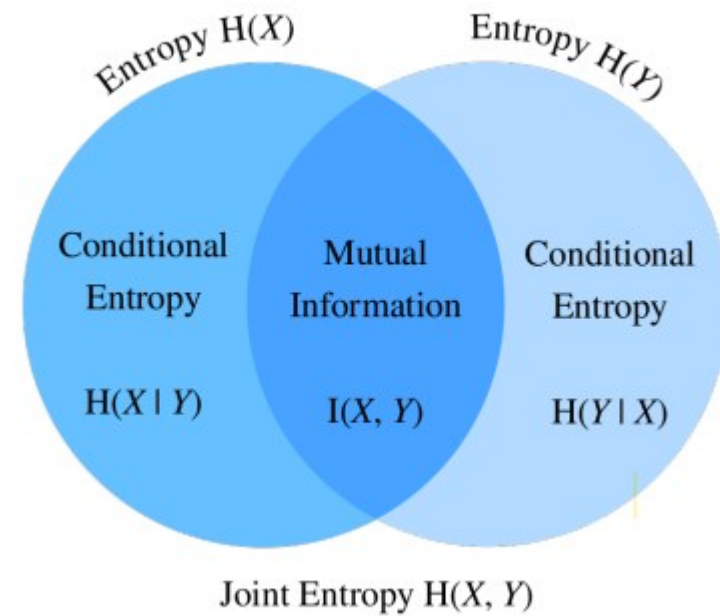
$$\begin{aligned}l(\theta) &= \log L(\theta) \\&= \log \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \\&= \sum_{i=1}^n y_i \log(\pi_i) + (1 - y_i) \log(1 - \pi_i).\end{aligned}$$

$$\text{CE}(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{i=1}^n y_i \log \hat{y}_i$$

Mutual Information

$$I(X, Y) = H(X, Y) - H(Y | X) - H(X | Y).$$

- $H(X) - H(X | Y)$
- $H(Y) - H(Y | X)$
- $H(X) + H(Y) - H(X, Y)$



Mutual Information

- Mutual information is symmetric, i.e., $I(X, Y) = I(Y, X)$.
- Mutual information is non-negative, i.e., $I(X, Y) \geq 0$.
- $I(X, Y) = 0$ if and only if X and Y are independent
- Alternatively, if X is an invertible function of Y , then Y and X share all information and

$$I(X, Y) = H(Y) = H(X).$$

KL (Kullback–Leibler) Divergence:

Kullback–Leibler (KL) divergence is a way to measure if two distributions are close together or not.

Given a random variable X that follows the probability distribution P with a p.d.f. or a p.m.f. $p(x)$, and we estimate P by another probability distribution Q with a p.d.f. or a p.m.f. $q(x)$. Then the Kullback–Leibler (KL) divergence (or relative entropy) between P and Q is:

$$D_{\text{KL}}(P\|Q) = E_{x \sim P} \left[\log \frac{p(x)}{q(x)} \right] .$$

We can interpret it as our relative surprise at observing the outcome compared to how surprised we would be observing it from our reference distribution.

Cross Entropy:

For a random variable X , we can also measure the divergence between the estimating distribution Q and the true distribution P via cross-entropy,

$$\text{CE}(P, Q) = -E_{x \sim P}[\log(q(x))].$$

$$\text{CE}(P, Q) = H(P) + D_{\text{KL}}(P \| Q).$$

1. Maximizing predictive probability of Q for distribution P , (i.e., $E_{x \sim P}[\log(q(x))]$);
2. Minimizing cross-entropy $\text{CE}(P, Q)$;
3. Minimizing the KL divergence $D_{\text{KL}}(P \| Q)$.

The definition of cross-entropy indirectly proves the equivalent relationship between objective 2 and objective 3, as long as the entropy of true data $H(P)$ is constant.

Summary

- Entropy is the unit to measure how much information is present in different signals.
- KL divergence can also measure the divergence between two distributions.
- Cross-entropy can be viewed as an objective function of multi-class classification.
- Minimizing cross-entropy loss is equivalent to maximizing the log-likelihood function.

Some Working Example

Property-1	Property-3	Property-2	Results
+	+	-	+
-	-	+	-
-	+	+	-
+	-	-	+
-	-	-	-
+	-	+	+
-	+	-	-

Some Working Example

Property-1	Property-3	Property-2	Results
+	+	-	+
-	-	+	-
-	+	+	-
+	-	-	+
-	-	-	-
+	-	+	+
-	+	-	-

Positive = 3 and Negative = 4

Some Working Example

Property-1	Property-3	Property-2	Results
+	+	-	+
-	-	+	-
-	+	+	-
+	-	-	+
-	-	-	-
+	-	+	+
-	+	-	-

$$\begin{aligned}\text{Entropy} &= [-3/7\log(3/7) - 4/7\log(4/7)] \\ &= 0.524 + 0.461 = 0.985\end{aligned}$$

Positive = 3 and Negative = 4

Some Working Example

Property-1	Property-3	Property-2	Results
+	+	-	+
-	-	+	-
-	+	+	-
+	-	-	+
-	-	-	-
+	-	+	-
-	+	-	-

$$\begin{aligned}\text{Entropy} &= [-2/7\log(2/7) - 5/7\log(5/7)] \\ &= 0.516 + 0.346 = 0.86\end{aligned}$$

Positive = 2 and Negative = 5

A Sample Joint Distribution

(x,y)	0	1
0	$1/2$	$1/4$
1	$1/8$	$1/8$

$$P(X|Y=1) = ?$$

$$P(Y|X=0) = ?$$

$$H(X,Y) = ?$$

$$H(X) = ?$$

$$H(Y) = ?$$

$$H(X|Y) = ?$$

$$H(Y|X) = ?$$

$$I(X,Y) = ?$$

A Sample Joint Distribution

x,y	0	1
0	1/2	1/4
1	1/8	1/8

$$P(X=0) = 5/8$$

$$P(X=1) = 3/8$$

$$P(Y=1) = 1/4$$

$$P(Y=0) = 3/4$$

$$P(X|Y=1) = ?$$

$$P(Y|X=0) = ?$$

$$H(X,Y) = ?$$

$$H(X) = ?$$

$$H(Y) = ?$$

$$H(X|Y) = ?$$

$$H(Y|X) = ?$$

$$I(X,Y) = ?$$

More Examples: Joint Probability Distribution

(X,Y)	0	1	Total
1	0.047	0.085	0.132
2	0.042	0.041	0.083
3	0.082	0.014	0.096
4	0.088	0.009	0.097
5	0.115	0.014	0.129
6	0.126	0.031	0.157
7	0.108	0.051	0.159
8	0.079	0.068	0.147
Total	0.687	0.313	1.000

Solution

$$P_X(X = x) = \sum_{\text{all } y} P_{XY}(x, y)$$

$$P_Y(Y = y) = \sum_{\text{all } x} P_{XY}(x, y)$$

$$E(X) = \sum_{\text{all } x} x P_X(x)$$

$$E(Y) = \sum_{\text{all } y} y P_Y(y)$$

$$Var(X) = \sum_{\text{all } x} x^2 P_X(x) - (E(X))^2$$

$$Var(Y) = \sum_{\text{all } y} y^2 P_Y(y) - (E(Y))^2$$

$$Cov(X, Y) = \sum_{\text{all } x} \sum_{\text{all } y} xy P_{XY}(x, y) - E(X)E(Y)$$

$$\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

$$P(X=1) = 0.132$$

$$P(X=2) = 0.083$$

$$P(X=8) = 0.147$$

$$P(Y=0) = 0.687$$

$$P(Y=1) = 0.313$$

$$E(X) = (1 \times 0.132) + (2 \times 0.083) + (3 \times 0.096) + (4 \times 0.097) + (5 \times 0.129) + (6 \times 0.157) + (7 \times 0.159) + (8 \times 0.147) = 4.85$$

$$E(Y) = (0 \times 0.687) + (1 \times 0.313) = 0.313$$

$$Var(X) = [(1^2 \times 0.132) + (2^2 \times 0.083) + (3^2 \times 0.096) + (4^2 \times 0.097) + (5^2 \times 0.129) + (6^2 \times 0.157) + (7^2 \times 0.159) + (8^2 \times 0.147)] - (4.85)^2 \Rightarrow 5.4335$$

$$Var(Y) = [(0^2 \times 0.687) + (1^2 \times 0.313)] - (0.313)^2 \Rightarrow 0.2150$$

Calculating the joint probabilities:

$P(X=1 Y=0) = 0.047$	$P(X=1 Y=1) = 0.085$
$P(X=2 Y=0) = 0.042$	$P(X=2 Y=1) = 0.041$
$P(X=3 Y=0) = 0.082$	$P(X=3 Y=1) = 0.014$
$P(X=4 Y=0) = 0.088$	$P(X=4 Y=1) = 0.009$
$P(X=5 Y=0) = 0.115$	$P(X=5 Y=1) = 0.014$
$P(X=6 Y=0) = 0.126$	$P(X=6 Y=1) = 0.031$
$P(X=7 Y=0) = 0.108$	$P(X=7 Y=1) = 0.051$
$P(X=8 Y=0) = 0.079$	$P(X=8 Y=1) = 0.068$

$P(X=1, Y=0) = P(X=1|Y=0) \times P(Y=0) = 0.047 \times 0.687$
 $P(X=2, Y=0) = P(X=2|Y=0) \times P(Y=0) = 0.042 \times 0.687$
 $P(X=3, Y=0) = P(X=3|Y=0) \times P(Y=0) = 0.082 \times 0.687$
 $P(X=4, Y=0) = P(X=4|Y=0) \times P(Y=0) = 0.088 \times 0.687$
 $P(X=5, Y=0) = P(X=5|Y=0) \times P(Y=0) = 0.115 \times 0.687$
 $P(X=6, Y=0) = P(X=6|Y=0) \times P(Y=0) = 0.126 \times 0.687$
 $P(X=7, Y=0) = P(X=7|Y=0) \times P(Y=0) = 0.108 \times 0.687$
 $P(X=8, Y=0) = P(X=8|Y=0) \times P(Y=0) = 0.079 \times 0.687$
 $P(X=1, Y=1) = P(X=1|Y=1) \times P(Y=1) = 0.085 \times 0.313$
 $P(X=2, Y=1) = P(X=2|Y=1) \times P(Y=1) = 0.041 \times 0.313$
 $P(X=3, Y=1) = P(X=3|Y=1) \times P(Y=1) = 0.014 \times 0.313$
 $P(X=4, Y=1) = P(X=4|Y=1) \times P(Y=1) = 0.009 \times 0.313$
 $P(X=5, Y=1) = P(X=5|Y=1) \times P(Y=1) = 0.014 \times 0.313$
 $P(X=6, Y=1) = P(X=6|Y=1) \times P(Y=1) = 0.031 \times 0.313$
 $P(X=7, Y=1) = P(X=7|Y=1) \times P(Y=1) = 0.051 \times 0.313$
 $P(X=8, Y=1) = P(X=8|Y=1) \times P(Y=1) = 0.068 \times 0.313$

$$\begin{aligned} \text{Cov}(X,Y) = & (0.047 \times 0.687)(-3.85)(-0.313) + (0.042 \times 0.687)(-2.85)(-0.313) + (0.082 \times 0.687)(-1.85)(-0.313) + (0.088 \times 0.687)(-0.85)(-0.313) \\ & + (0.115 \times 0.687)(0.15)(-0.313) + (0.126 \times 0.687)(1.15)(-0.313) + (0.108 \times 0.687)(2.15)(-0.313) + (0.079 \times 0.687)(3.15)(-0.313) \\ & + (0.085 \times 0.313)(-3.85)(0.687) + (0.041 \times 0.313)(-2.85)(0.687) + (0.014 \times 0.313)(-1.85)(0.687) + (0.009 \times 0.313)(-0.85)(0.687) \\ & + (0.014 \times 0.313)(0.15)(0.687) + (0.031 \times 0.313)(1.15)(0.687) + (0.051 \times 0.313)(2.15)(0.687) + (0.068 \times 0.313)(3.15)(0.687) = -0.0499 \end{aligned}$$

$$\text{Correlation} = \frac{-0.0499}{\sqrt{5.4}\sqrt{0.21}} \Rightarrow -0.046$$

References and Further Reading

- Zhang, Aston, Zachary C. Lipton, Mu Li, and Alexander J. Smola. "Dive into deep learning." *arXiv preprint arXiv:2106.11342* (2021).
- Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong